Jongwoo Kim

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RESEARCH AREA

1) World Models and JEPA: bringing latent predictive learning to real domains such as pathology, graphs, and education. 2) Computational Pathology: modeling the hierarchical, multi-scale structure of gigapixel slides, from local tissue patterns to the whole slide. 3) Graph Learning: capturing the right inductive biases for graph-structured data, so that models generalize across graphs.

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** *Aug. 2022 – Present*
  Ph.D. in Industrial and Systems Engineering; M.S. in Data Science Daejeon, South Korea
  - **Ph.D. in Industrial and Systems Engineering** (Aug. 2024 – Present)
    - Advisors: Prof. Mun Yong Yi and Prof. Noseong Park
  - **M.S. in Data Science** (Sep. 2022 – Aug. 2024)
    - Thesis: *MicroMIL: Spatially Enhanced Multiple Instance Learning for Advanced Cancer Diagnosis in Microscopy*
    - Advisor: Prof. Mun Yong Yi
- **Seoul National University of Science and Technology** *Feb. 2016 – Aug. 2022*
  B.S. in Industrial and Systems Engineering (Summa Cum Laude) Seoul, South Korea
  - Thesis: *How to succeed in the market? Predicting startup success using a machine learning approach*
  - Advisor: Prof. Youngjung Geum

CONFERENCE AND JOURNAL PUBLICATIONS

C = CONFERENCE, J = JOURNAL

† *Equal contribution.*

[C.14] **Jongwoo Kim**, Seongyeub Chu, Jeongwhan Choi, Jiwon Jeong, Mun Yi. **Knowledge World Model for Learning Path Recommendation.** In *CIKM*, 2026.

[C.13] Seongyeub Chu, **Jongwoo Kim**, Mun Yi. **FeedEval: Pedagogically Aligned Evaluation of LLM-Generated Essay Feedback.** In *Findings of ACL*, 2026.

[C.12] Jeongwhan Choi†, **Jongwoo Kim**†, Woosung Kang, Noseong Park. **Learning Posterior Predictive Distributions for Node Classification from Synthetic Graph Priors.** In *ICLR*, 2026.

[C.11] Bryan Wong†, **Jongwoo Kim**†, Huazhu Fu, Mun Yi. **Few-Shot Learning from Gigapixel Images via Hierarchical Vision-Language Alignment and Modeling.** In *NeurIPS*, 2025.

[C.10] **Jongwoo Kim**†, Seongyeub Chu†, Bryan Wong, Mun Yi. **Not All Options Are Created Equal: Textual Option Weighting for Token-Efficient LLM-Based Knowledge Tracing.** In *Findings of EMNLP*, 2025.

[C.9] **Jongwoo Kim**, Seongyeub Chu, Hyeongmin Park, Bryan Wong, Keejun Han, Mun Yi. **Leveraging Multi-facet Paths for Heterogeneous Graph Representation Learning.** In *CIKM*, 2025.

[C.8] Chanjae Song†, Seongyeub Chu†, **Jongwoo Kim**†, Mun Yi. **A Novel Evaluation Framework for 15-Minute City Using Satellite Imagery.** In *ACM SIGSPATIAL*, 2025.

[C.7] **Jongwoo Kim**†, Bryan Wong†, Huazhu Fu, Willmer Rafell Quiñones Robles, Mun Yi. **MicroMIL: Graph-Based Multiple Instance Learning for Context-Aware Diagnosis with Microscopic Images.** In *MICCAI*, 2025.

[C.6] Seongyeub Chu†, **Jongwoo Kim**†, Bryan Wong, Mun Yi. **Rationale Behind Essay Scores: Enhancing S-LLM's Multi-Trait Essay Scoring with Rationale Generated by LLMs.** In *Findings of NAACL*, 2025.

[C.5] Seongyeub Chu†, **Jongwoo Kim**†, Mun Yi. **Think Together and Work Better: Combining Humans' and LLMs' Think-Aloud Outcomes for Effective Text Evaluation.** In *CHI*, 2025. **Oral**.

[C.4] Hyeongmin Park, Sungrae Hong, Chanjae Song, **Jongwoo Kim**, Mun Yi. **Uncertainty-Based Data-Wise Label Smoothing for Calibrating Multiple Instance Learning in Histopathology Image.** In *WACV*, 2025. **Oral**.

[C.3] **Jongwoo Kim**, Minkyu Jeong, Hyeongmin Park, Young Sin Ko, Mun Yi. **Leveraging Spatial Relationships in Microscopic Images for Patient Cancer Diagnosis.** In *ISBI*, 2024.

[C.2] Seongyeub Chu, **Jongwoo Kim**, SeungTai Yoo, Mun Yi. **Option-Knowledge Concept Aware Knowledge Tracing: Modeling Student Proficiency with Process-Oriented Assessment in Education.** In *Korea Computer Congress (KCC)*, 2024.

[J.2] **Jongwoo Kim**, Hongil Kim, Youngjung Geum. **How to Succeed in the Market? Predicting Startup Success Using a Machine Learning Approach.** *Technological Forecasting & Social Change (TFSC)*, 2023. (IF = 13.3)

[C.1] Sungrae Hong, Taemi Kim, Sol Lee, **Jongwoo Kim**, Mun Yi. **MATE: The Multimodal Model Using Audio and Text for Emotion Recognition.** In *Korea Computer Congress (KCC)*, 2022.

[J.1] Mingyu Park, Byeongmin Jeon, **Jongwoo Kim**, Youngjung Geum. **Exploring Potential Application Industry for Fintech Technology by Expanding Its Terminology: Network Analysis and Topic Modelling Approach.** *The Journal of Society for e-Business Studies*, 2021.

Papers Under Review and Preprints

S = Submitted / Under Review

† *Equal contribution.*

[S.11] Jeongwhan Choi, Jonghyeok Shin, **Jongwoo Kim**, Woosung Kang, Noseong Park. **Looping in Context: Node Classification with Looped Prior-Fitted Networks.** Under Review, 2026.

[S.10] Woosung Kang, Jeongwhan Choi, **Jongwoo Kim**, Donghwan Shin, Noseong Park. **LPLP: Learning Patterns for Link Prediction from Context.** Under Review, 2026.

[S.9] Hyunji Kim†, **Jongwoo Kim**†, Chanjae Song, Mun Yi. **LaViM: Latent Visual Memory for Multi-Agent Whole Slide Image Reasoning.** Under Review, 2026.

[S.8] **Jongwoo Kim**, Chanjae Song, Jeongwhan Choi, Woosung Kang, Noseong Park, Mun Yi. **Learning Graph Representations via Predictive Language Supervision.** Under Review, 2026.

[S.7] **Jongwoo Kim**, Willmer Rafell Quiñones Robles, Bryan Wong, Mun Yi. **Cross-Scale Latent Predictive Pretraining for Whole-Slide Image.** Under Review, 2026.

[S.6] **Jongwoo Kim**, Jeongwhan Choi, Seongyeub Chu, Noseong Park, Mun Yi. **GraPE: Graph Prompt Tuning via Mixture of Experts Approach.** Under Review, 2026.

[S.5] Chanjae Song, **Jongwoo Kim**, Mun Yi. **Decoupled Graph Experts for Robust Multimodal Learning with Missing Modalities.** Under Review, 2026.

[S.4] Jeongwhan Choi, Woosung Kang, Minseo Kim, **Jongwoo Kim**, Noseong Park. **Can TabPFN Compete with GNNs for Node Classification via Graph Tabularization?** Preprint, 2025.

[S.3] Willmer Rafell Quiñones Robles, Sakonporn Noree, **Jongwoo Kim**, Young Sin Ko, Bryan Wong, Mun Yi. **Leveraging Spatial Context for Positive Pair Sampling in Histopathology Image Representation Learning.** Preprint, 2025.

[S.2] Seongyeub Chu, **Jongwoo Kim**, Mun Yi. **AMR: Aspect-Aware MOOC Recommendation in a Heterogeneous Network.** Preprint, 2026.

[S.1] **Jongwoo Kim**, Seongyeub Chu, Mun Yi. **RFMRec: Enhancing Personal Recommendation Using Public Trend Captured via RFM.** Preprint, 2025.

Projects

- **Semiconductor Defect Detection with ETRI & Bluetile Lab (Project Manager)** *Sep. 2025 – Present*
  Tools: Vision Transformer, Weakly Supervised Learning
  ◦ Developed a defect detection pipeline for semiconductor wafer images with **ETRI** and **Bluetile Lab**.
- **Large Vision-Language Model for Pathology Diagnosis with NPF (Project Manager)** *Sep. 2025 – Present*
  Tools: Vision Language Model, Human Computer Interaction
  ◦ Developed a Human-Centered VLM pipeline for histopathology.
- **Personalized Problem Recommendation with Chunjae Education** *Feb. 2025 – Dec. 2025*
  Tools: Knowledge Tracing, Graph Embedding
  ◦ Built a personalized problem recommendation system using **Knowledge Tracing** and **Transition Graph**.
- **Multi-view Essay Scoring with KT Corporation** *Jul. 2024 – Dec. 2024*
  Tools: Transformers, BERTScore
  ◦ Designed a **multi-view essay scoring model** integrating linguistic, semantic, and stylistic features with **KT**.
- **Colon Cancer Diagnosis with Seegene Medical Foundation (Project Manager of Microscope)** *Jul. 2023 – Jul. 2024*
  Tools: Multiple Instance Learning, GNN
  ◦ Developed a **graph-based MIL model** for colon cancer diagnosis using whole-slide microscopy images.
- **Seoul Real Estate Price Prediction with Tushin** *Feb. 2023 – Jul. 2023*
  Tools: XGBoost, LightGBM, SHAP, Geospatial Analysis
  ◦ Built a **real estate price prediction model** using public datasets and transaction records in collaboration with **Toojae-shin**.

Honors and Awards

- **Kim Young-Han Global Leader Scholarship** *2025*
  KAIST, Korea
- **Ph.D. Student Research Encouragement Grant** *Sep. 2025 – Aug. 2027*
  National Research Foundation of Korea (NRF), Korea
- **Industrial and Social Issues Poster Contest (3rd Place)** *Nov. 2025*
  KAIST, Korea
- **Best Paper Award** *Jun. 2024*
  Korea Computer Congress (KCC), Korea
- **Industrial and Social Issues Poster Contest (2nd Place)** *Nov. 2022*
  KAIST, Korea
- **ETRI Human Understanding Artificial Intelligence Paper Contest (Minister of Science and ICT Award)** *Jun. 2022*
  Electronics and Telecommunications Research Institute (ETRI), Korea
- **Capstone Design (2nd Place)** *Dec. 2021*
  Seoul National University of Science and Technology, Korea
- **Superior Academic Performance** *Aug. 2020 – Feb. 2022*
  Seoul National University of Science and Technology, Korea

Professional Services

- **Conference Reviewer**
  - ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2024, 2026
  - International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2025, 2026
  - The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026